column-hse-z: hydrostatic columns on a height grid

The physics of `run.mode = "column-hse-z"`, explained step by step (B.Sc. level). HSEMHS.

1. The question

We are given the temperature T of a piece of the solar atmosphere at every point of a three-dimensional grid: two horizontal coordinates x, y and the height z. We want to know the **gas pressure** P and the **density** ρ at every point, and from them how transparent the gas is, i.e. the **optical depth** τ_{500} at the wavelength 500 nm. The answer this mode gives is the simplest physically consistent one: every vertical column of the grid is treated as a separate, static column of gas that carries its own weight. Nothing moves, and neighbouring columns do not interact.

2. The quantities

Symbol	Meaning	Unit
x, y, z	horizontal coordinates and height (z grows upward; the grid stores the top level first)	m
T	temperature, given, never changed	K
P	gas pressure: force per area exerted by the gas	$\mathrm{Pa} = \mathrm{N\,m^{-2}}$
u	$\ln(P/1\,\mathrm{Pa})$, the natural logarithm of the pressure	–
ρ	mass density	$\mathrm{kg\,m^{-3}}$
g	gravitational acceleration at the solar surface, $g = 274$	$\mathrm{m\,s^{-2}}$
n	number of free particles (atoms, ions, electrons, molecules) per volume	$\mathrm{m^{-3}}$
n_e, P_e	number density of free electrons; electron pressure $n_e k_B T$	$\mathrm{m^{-3}}$, Pa
k_B	Boltzmann constant, 1.381×10^{-23}	$\mathrm{J\,K^{-1}}$
m_H	mass of a hydrogen atom, 1.67×10^{-27}	kg
$\bar{\mu}$	mean molecular weight: average mass of a free particle in units of m_H	–
H	pressure scale height	m
m	column mass: the mass of gas above a point, per unit horizontal area	$\mathrm{kg\,m^{-2}}$
P_ref	the pressure prescribed at one level of every column (the anchor)	Pa
χ_{500}	extinction coefficient at 500 nm: fraction of the light removed per metre	$\mathrm{m^{-1}}$
τ_{500}	optical depth at 500 nm, counted from the top downward	–

3. Hydrostatic equilibrium

Take a thin horizontal slab of gas with area A, between heights z and $z + dz$. Three vertical forces act on it:

- the gas below pushes it up with the force $P(z)\,A$;
- the gas above pushes it down with $P(z + dz)\,A$;
- gravity pulls it down with its weight, mass $\times\, g = \rho\,A\,dz\,g$.

If the slab does not accelerate, the forces cancel: $P(z)A - P(z+dz)A - \rho g A\,dz = 0$. Dividing by $A\,dz$ and letting $dz \to 0$:

$$\boxed{\frac{dP}{dz} = -\rho\,g}$$

The pressure *decreases* upward, and it decreases fastest where the gas is densest. This is **hydrostatic equilibrium (HSE)**. It is the same law that makes the air pressure fall as you climb a mountain.

4. Equation of state: how ρ follows from T and P

4.1 The ideal gas

The solar photosphere and chromosphere behave as an ideal gas: $P = n\,k_B T$. If the free particles have an average mass $\bar{\mu}\,m_\mathrm{H}$, the density is $\rho = n\,\bar{\mu}\,m_\mathrm{H}$, so

$$\rho = \frac{\bar{\mu}\,m_\mathrm{H}}{k_B}\,\frac{P}{T}.$$

At fixed temperature, doubling the pressure doubles the density.

4.2 Why $\bar{\mu}$ is not a constant

For neutral solar gas (about 91 % hydrogen and 9 % helium atoms by number) $\bar{\mu} \approx 1.3$. When hydrogen ionises, each atom becomes two particles, a proton and an electron, with almost the same total mass, so $\bar{\mu}$ roughly halves

(to ≈ 0.6 for fully ionised gas). In the coolest layers hydrogen forms H_2 molecules, which raises $\bar{\mu}$. How much is ionised is described by the **Saha equation**, for an atom in ionisation stage i:

$$\frac{n_{i+1}\, n_e}{n_i} = \frac{2U_{i+1}}{U_i} \left(\frac{2\pi m_e k_B T}{h^2} \right)^{3/2} e^{-E_i/k_B T},$$

where E_i is the ionisation energy (13.6 eV for hydrogen), U are partition functions and h is Planck's constant. Two features matter. The exponential makes ionisation switch on rapidly with temperature. And the factor n_e on the left means that at higher density (more electrons around to recombine with) a higher temperature is needed for the same ionisation. So $\bar{\mu}$ depends on *both* T and P.

4.3 The table

Solving the Saha equations for all relevant elements together with charge conservation and molecule formation is too slow to do at every point of every step. It is done once, with Wittmann's equation-of-state code for solar composition, on a fine grid of $(\ln T, \ln P)$, and stored in a **table**. The code reads $\ln \rho$, $\ln n_e$ and the 500 nm extinction $\ln \chi_{500}$ from it by smooth (bicubic) interpolation. Because all three come from the same solution, they are consistent with each other.

5. The scale height and why we work with $\ln P$

Insert the equation of state into hydrostatic equilibrium:

$$\frac{dP}{dz} = -\frac{\bar{\mu} m_{\mathrm{H}} g}{k_B T} P = -\frac{P}{H}, \qquad H \equiv \frac{P}{\rho g} = \frac{k_B T}{\bar{\mu}\, m_{\mathrm{H}}\, g}.$$

H, the **pressure scale height**, has the dimension of a length. For $T = 6000$ K and $\bar{\mu} = 1.3$ it is

$$H = \frac{1.38 \times 10^{-23} \cdot 6000}{1.3 \cdot 1.67 \times 10^{-27} \cdot 274}\ \mathrm{m} \approx 1.4 \times 10^5\ \mathrm{m} = 140\ \mathrm{km}.$$

If H were constant (an isothermal gas), the solution would be $P(z) = P(z_0)\, e^{-(z-z_0)/H}$: the pressure falls by a factor $e \approx 2.7$ every 140 km. In general H changes with height, and dividing by P gives the form the code uses:

$$\boxed{\frac{du}{dz} = -\frac{1}{H(T,P)}, \qquad u = \ln P} \qquad \Rightarrow \qquad u(z) = u(z_0) - \int_{z_0}^{z} \frac{dz'}{H(z')}.$$

Working with $u = \ln P$ has two advantages. A column spans many factors of ten in pressure, but u changes smoothly and by moderate amounts, which is what a numerical integration handles best. And $P = e^u$ is automatically positive.

6. The anchor: one pressure per column

$du/dz = -1/H$ is a first-order differential equation: it determines how u changes, not its value. Like any such equation it needs one starting value per column. The mode prescribes the **same pressure P_{ref} at one level in every column**, by default the bottom level (`anchor = "bottom"`), and integrates upward from there. The other options are the top level (integrate downward) and a chosen height z_{ref} inside the grid (integrate both ways).

Why the bottom? Deep in the atmosphere the gas is dense and the temperature differences between columns matter least, so a common pressure there is a reasonable approximation. A top anchor would make the pressure in the photosphere depend on the very different temperatures of the chromosphere and transition region in each column. Since every column starts from the same P_{ref}, all differences between columns come from their temperatures.

Between grid levels the temperature is not known. The code assumes that $\ln T$ varies linearly with z inside each interval and integrates with the classical fourth-order Runge–Kutta method, which evaluates the slope $-1/H$ four times per step and averages them. Each interval is computed with n and $2n$ substeps; if the two disagree by more than 10^{-6} in u, the number of substeps is doubled until they agree. Steep temperature changes (the transition region) thus get more substeps automatically.

7. Column mass

Integrating $dP/dz = -\rho g$ from a height z up to the top of the column gives

$$P(z) - P_{\mathrm{top}} = g \int_z^{z_{\mathrm{top}}} \rho\, dz' = g\, m(z),$$

where $m(z)$ is the mass of gas above z per square metre. *The pressure at any point is the weight of the gas above it* (plus whatever presses on the top). The code integrates m along with u and stores it, which also serves as a check of the integration.

8. Light, opacity and optical depth

8.1 Extinction and the Beer–Lambert law

Light of intensity I travelling a distance ds through gas loses a fraction $\chi\, ds$:

$$dI = -\chi\, I\, ds \qquad \Rightarrow \qquad I(s) = I(0)\, e^{-\int_0^s \chi\, ds'} = I(0)\, e^{-\tau}.$$

The number $\tau = \int \chi\, ds$, the **optical depth**, counts how many "absorption lengths" the light crosses. χ_{500} includes both true absorption and scattering at 500 nm. In the solar photosphere it is dominated by the negative hydrogen ion H^- (a hydrogen atom with an extra electron, bound by only 0.75 eV). Forming H^- needs free electrons, which come mostly from easily ionised metals (Na, Ca, Fe, ...), so χ_{500} rises steeply with temperature and with density. It comes from the same table as ρ.

8.2 Optical depth along the vertical

For light leaving the Sun vertically, the optical depth of a point at height z is

$$\tau_{500}(z) = \tau_{\text{top}} + \int_z^{z_{\text{top}}} \chi_{500}(z')\, dz'.$$

It is zero at the top of the atmosphere and grows downward. Light emitted at optical depth τ escapes with probability $e^{-\tau}$, so most of the light we see comes from around $\tau \approx 1$ (a result known as the Eddington–Barbier relation: the emerging intensity is close to the emission at $\tau = 1$). The height $z_{\tau 1}(x, y)$ where $\tau_{500} = 1$ is the "visible surface" of the Sun. Spectral lines form at fixed optical depth, so the τ scale is what an observation or an inversion actually probes. For a viewing angle with $\mu = \cos\theta$ the path is longer by $1/\mu$, and $\tau_\mu = \tau/\mu$.

8.3 The gas above the box

The grid ends at z_{top}, but the atmosphere does not. If χ continues to fall exponentially above the top, $\chi(z) = \chi_1 e^{-(z-z_1)/H_\chi}$, the missing optical depth is

$$\tau_{\text{top}} = \int_{z_1}^{\infty} \chi_1 e^{-(z-z_1)/H_\chi}\, dz = \chi_1 H_\chi, \qquad H_\chi = \frac{z_1 - z_2}{\ln(\chi_2/\chi_1)},$$

with H_χ taken from the top two levels (index 1 is the top). If χ does not decrease upward there, this makes no sense, and half a grid cell is used instead, $\tau_{\text{top}} = \frac{1}{2}\chi_1(z_1 - z_2)$.

9. What the answer means, and what it does not

Each column is now in exact vertical equilibrium. But consider two neighbouring columns at the same height, one hot and one cool. Their pressures differ, so there is a horizontal pressure gradient $\partial P/\partial x \neq 0$: a sideways force that nothing balances. The atmosphere is **not** in three-dimensional equilibrium. The mode reports this horizontal force as a diagnostic and labels its output *vertical HSE only.*

Could any choice of pressure make a non-magnetic atmosphere fully static? The full force balance is $-\nabla P - \rho g \hat{\mathbf{z}} = 0$. Take the curl of both sides. The curl of a gradient is always zero, and $\nabla \times (\rho g \hat{\mathbf{z}}) = g\,(\partial_y \rho,\, -\partial_x \rho,\, 0)$, so

$$\frac{\partial \rho}{\partial x} = \frac{\partial \rho}{\partial y} = 0.$$

A static atmosphere without magnetic field must have the same density everywhere on each horizontal plane, and then the horizontal equations give $\partial_x P = \partial_y P = 0$ too. A real, structured atmosphere (hot granules next to cool lanes) cannot be static without help: in the Sun that help is the motion of the gas, and in a static model it can only be the magnetic field. That is the subject of the `mhs-*` and `full-mhs-z` modes, which use this mode's result as their starting point.